CS3230 Lecture 2

"Asymptotics, Summation in Analysis of Algorithms"

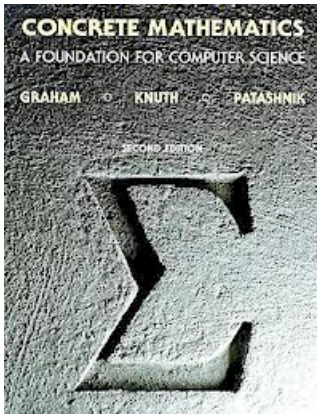
- **Lecture Topics and Readings**
  - **Simple Analysis of Algorithms (AA-P)** **[CLRS]-C2**
  - **Asymptotic Notations O, Θ, Ω, o, ω,** **[CLRS]-C3.1**
  - **Summation and Bounding** **[CLRS]-App.A**
- **Assignment:** HW1: Fri, 24-Jan (< 12nn)

Algorithms & Discrete Math
have very happy, stable marriage!

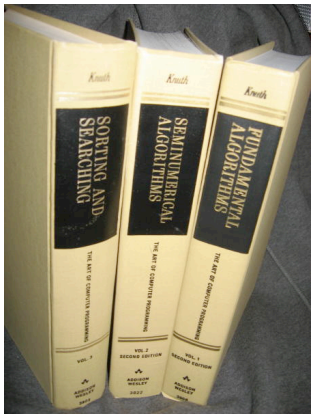
Hon Wai Leong, NUS
© Leong Hon Wai, 2003--

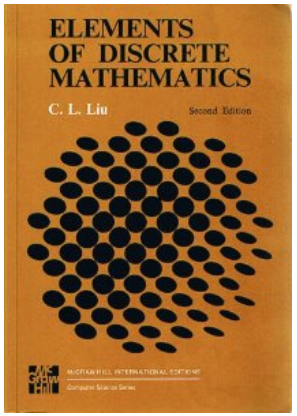
Early pioneers of Math for CS

Don Knuth
Stanford



C. L. Liu,
MIT, UIUC,
NTHU (tw)





© Leong Hon Wai, 2003--

Visiting Dave in Taipei (Feb 2006)

© Leong Hon Wai, 2003--

CS3230 Announcements

- **Read "Cattywampus"**
- **Read "Remarks about Homework"**
- **Read "CS3230 Announcements"**
- HW1 due this Friday (< 12nn)
- Tutorial starts new week

© Leong Hon Wai, 2003--

Surprise Quiz 1

Q1: When is HW1 due?

a) Next semester

b) Next month (after CNY)

c) Huh? “What do you mean HW?

d) Next week (Friday, < 12nn)

Q2: Where to turn in HW1 answers?

a) NOT, IVLE workbin

b) NOT to Undergrad Office

c) NOT to my house

d) To the CS3230 Dropbox (outside my office)

© Leong Hon Wai, 2003--

Tutorial Grouping:

- So far, 77 (of 90) has signed up
- Need to achieve better balance! 90/6=15
  - Move away from G1 … to G2-4, G6

The numbers in diff groups:

- G1: 20 G4: 10
- G2: 10 G5: 16
- G3: 13 G6: 8

© Leong Hon Wai, 2003--

Programming Assignment

- **Programming Assignment 1:**
  - **Subtasks A,B,C (simple tasks)**
  - **Will be released on Monday, 27-Jan**
- **Will send you**
  - **userid+password, a template trial-program,**
- **You should *quickly***
  - **Log in to server; Do a trial-submit**
  - **Trial-program (does not work), but**
  - **Will get respond from the Server.**
- **Can work on problem right away**

© Leong Hon Wai, 2003--

Thank you.

Q & A

© Leong Hon Wai, 2003--

© Leong Hon Wai, 2003--

Chessboards, Hats, and Poetry : Some Rigorous and Not-So-Rigorous Mathematical Results

數裡有詩？ 詩裡有數！

C. L. Liu　　劉炯朗